PHYS101/Summer 2020
FINAL EXAM

(On my honor I have neither given nor received any unauthorized assistance on this exam.)

PHYSICS 101 NO: _____

NAME: _____

- This is a closed-everything exam.
- Calculators or computing devices are neither allowed nor needed.
- Exam duration is **90 minutes**.
- Consists of 6 problems worth of **100 points**.
- Answers without work shown or proper explanation will not be given any credit.

Prob. 1	Prob. 2	Prob. 3	Prob. 4	Prob. 5	Prob. 6	Total

Formula Sheet

Vector Products:
Scalar Product
$\vec{A}\cdot\vec{B} = |\vec{A}||\vec{B}|\cos\theta$
Vector Product
$|\vec{A}\times\vec{B}| = |\vec{A}||\vec{B}|\sin\theta$

Constant x-acceleration:
$v_x = v_{0x} + a_x t$
$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2$
$v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$
$x - x_0 = \left(\frac{v_{0x}+v_x}{2}\right)t$

Uniform Circular motion:
$a_{\text{rad}} = \frac{v^2}{R} \quad T = \frac{2\pi R}{v}$

Newton's 2nd Law:
$\sum \vec{F} = m\vec{a}$

Friction Force:
$f_k = \mu_k n \quad f_s \le \mu_s n$

Work done by a force:
$W = \vec{F}\cdot\vec{s} = Fs\cos\phi$

Kinetic energy:
$K = \frac{1}{2}mv^2$

The work-energy theorem:
$W_{\text{tot}} = K_2 - K_1 = \Delta K$

Work done by varying force:
$W = \int_{\vec{x}_1}^{\vec{x}_2} \vec{F}\cdot d\vec{x}$

Power:
$P_{\text{av}} = \frac{\Delta W}{\Delta t} \quad , \quad P_{\text{av}} = \vec{F}\cdot\vec{v}$

Gravitational Potential E.:
$\Delta U_g = -W_{\text{grav}} = mgy_f - mgy_i$

Elastic pot. energy:
$\Delta U_{el} = -W_{\text{el}} = \frac{1}{2}kx_f^2 - \frac{1}{2}kx_i^2$

Conservation of mech. energy:
$K_1 + U_1 + W_{\text{other}} = K_2 + U_2$
or: $\Delta K + \Delta U = W_{\text{other}}$

Momentum of a particle:
$\vec{p} = m\vec{v} \quad , \quad \sum \vec{F} = \frac{d\vec{p}}{dt}$

Impulse & Impulse-Momentum:
$\vec{J} = \sum \vec{F}\Delta t \quad , \quad \vec{J} = \Delta\vec{p} = \vec{p}_2 - \vec{p}_1$

Impulse of a varying force:
$\vec{J} = \int \vec{F}dt$

Conservation of momentum:
$\vec{P} = \vec{p}_A + \vec{p}_B + \dots$
$\vec{P}_i = \vec{P}_f \ (\vec{P} = \vec{P}')$

Elastic collision:
in case of $v_{B1x} = 0$:
$v_{A2x} = \frac{m_A - m_B}{m_A + m_B}v_{A1x}$
$v_{B2x} = \frac{2m_A}{m_A + m_B}v_{A1x}$

Rotational Kinematics:
$\omega_z = \frac{d\theta}{dt} \quad , \quad \alpha_z = \frac{d\omega_z}{dt}$
$\theta = \theta_0 + \omega_{0_z}t + \frac{1}{2}\alpha_z t^2$
$\theta - \theta_0 = \frac{1}{2}(\omega_{0_z} + \omega_z)t$

Relating Linear and Angular Kinematics (Rolling without Slipping/Sliding):
$v = r\omega \quad , \quad a_{\tan} = r\alpha$
$a_{\text{rad}} = \frac{v^2}{r} = \omega^2 r$

Moment of Inertia and Rotational Kinetic Energy:
$I = \sum_i m_i r_i^2 \quad , \quad K = \frac{1}{2}I\omega^2$

Parallel Axis Theorem:
$I_P = I_{\text{cm}} + Md^2$

Torque and Rotational Dynamics:
$\vec{\tau} = \vec{r}\times\vec{F} \quad , \quad \sum \tau_z = I\alpha_z$

Combined Tranlation and Rotation:
$\sum \vec{F}_{\text{ext}} = M\vec{a}_{\text{cm}} \quad , \quad \sum \tau_z = I_{\text{cm}}\alpha_z$
$v_{\text{cm}} = R\omega$ (rolling without slipping)
$K = \frac{1}{2}Mv_{\text{cm}}^2 + \frac{1}{2}I_{\text{cm}}\omega^2$

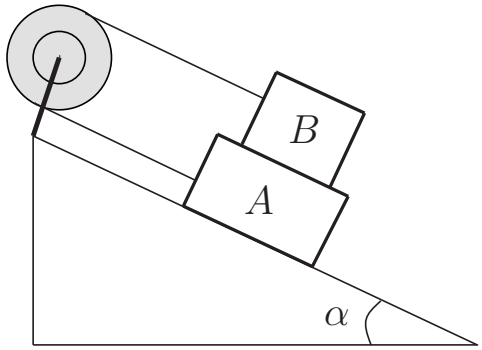
Work Done by a Torque:
$W_\tau = \tau_z\Delta\theta$ (const. torque)
$W_\tau = \int_{\theta_1}^{\theta_2} \tau_z d\theta \quad , \quad P = \tau_z\omega_z$

Work-Energy Theorem in Rotation:
$W_\tau = \Delta K_{rot} = \frac{1}{2}I\omega_2^2 - \frac{1}{2}I\omega_1^2$

Angular Momentum:
$\vec{L} = \vec{r}\times\vec{p} = \vec{r}\times m\vec{v}$ (for a particle)
$\vec{L} = I\vec{\omega}$ (for a rotating rigid body)

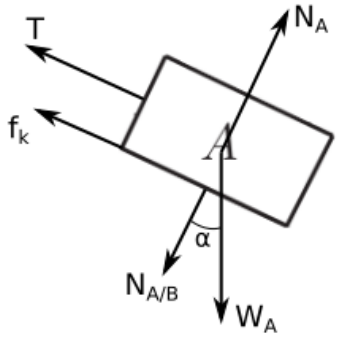
Rotational Dynamics and Angular Momentum:
$\sum \vec{\tau} = \frac{d\vec{L}}{dt}$
$\sum \vec{\tau} = 0 \Rightarrow \vec{L}$: constant $(\vec{L}_i = \vec{L}_f)$

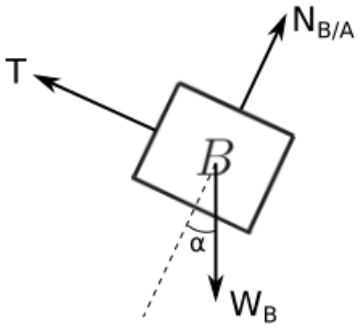
Problem 1 (15 points)



Two blocks with weights $w_A = 18$ N and $w_B = 12$ N, connected by a light rope passing through a frictionless pulley, are placed on an inclined surface ($\alpha = 45°$) as shown in the figure. The block A moves with a constant velocity in downward direction, and there is no friction between block A and B.

(a) (5pts.) Draw free-body diagrams for each block.





(b) (10pts.) Find the magnitude of the friction coefficient (μ) between the block A and surface.

Since blocks move with constant velocity, there is no acceleration. Apply Newton's 1st law:

For B:

$$\sum F_x = T - W_B \sin\alpha = 0 \Longrightarrow T = m_B g \sin\alpha$$

$$\sum F_y = N_{B/A} - W_B \cos\alpha = 0 \Longrightarrow N_{B/A} = m_B g \cos\alpha$$

For A:

$$\sum F_y = N_A - N_{A/B} - W_A \cos\alpha = 0 \Longrightarrow N_A = N_{A/B} + m_A g \cos\alpha$$

$$\sum F_x = W_A \sin\alpha - T - f_k = 0 \Longrightarrow f_k = m_A g \sin\alpha - T$$

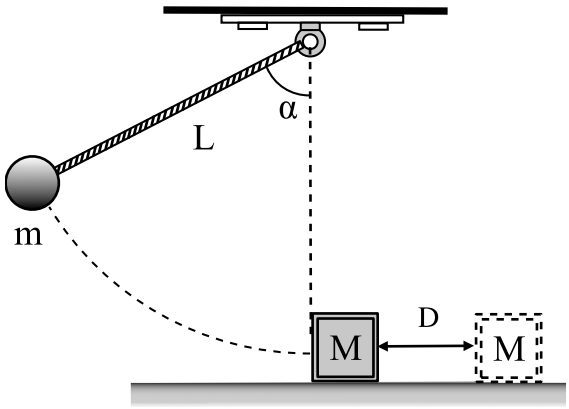
$$f_k = N_A \mu = m_A g \sin\alpha - T \Longrightarrow \mu = \frac{m_A g \sin\alpha - m_B g \sin\alpha}{N_{A/B} + m_A g \cos\alpha}$$

Since $N_{A/B} = N_{B/A}$,

$$\mu = \frac{m_A g \sin\alpha - m_B g \sin\alpha}{m_B g \cos\alpha + m_A g \cos\alpha} = \frac{g \sin\alpha (m_A - m_B)}{g \cos\alpha (m_A + m_B)} = \tan\alpha \frac{m_A - m_B}{m_A + m_B} = \tan 45° \left(\frac{18\text{kg} - 12\text{kg}}{18\text{kg} + 12\text{kg}}\right)$$

$$\Longrightarrow \boxed{\mu = 0.2}$$

Problem 2 (15 points)



A pendulum consisting of a ball of mass m and a massless string of length $L = 10.00$ m is released from an angle of $\alpha = 60°$ as shown in the figure and elastically strikes a block of mass $M = 3m$. The block slides a distance of 12.5m before stopping under the action of a constant friction force with the friction constant μ_k.
Hints: Take $g = 10$ m/s^2. $\sin\ 60° \simeq 0.87$, $\cos\ 60° \simeq 0.5$.

(a) (9 pts.) Find the radial acceleration of the ball <u>just after</u> the pendulum hits the block.

Apply conservation of mechanical energy for the ball:

$$\Delta K + \Delta U = 0 \Longrightarrow \frac{1}{2}mv^2 - mgL(1-\cos\alpha) = 0 \Longrightarrow v = \sqrt{2gL(1-\cos\alpha)} \Longrightarrow v = 10\text{m/s}$$

Apply conservation of momentum and kinetic energy for the ball and block: (v_p for the pendulum after collision and v_b for the block after collision)

$$v_p = \frac{m_p - m_b}{m_p + m_b}v = \frac{m-3m}{m+3m}(10\text{m/s} = -5\text{m/s})$$

$$v_b = \frac{2m_p}{m_p + m_b}v = \frac{2m}{m+3m}(10\text{m/s}) = 5\text{m/s}$$

$$a_{\text{rad}} = \frac{v_p^2}{L} = \frac{(-5\text{m/s})^2}{10\text{m}} \Longrightarrow \boxed{a_{\text{rad}} = 2.5\text{rad/s}}$$

(b) (6 pts.) Find the friction coefficient μ_k.

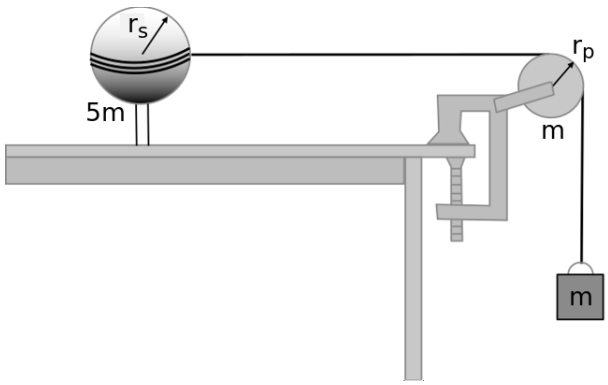
Apply conservation of total energy for the block:

$$\Delta K + \Delta U = W_d \Longrightarrow -\frac{1}{2}mv_b^2 = \vec{f}_k \cdot \vec{s} = -n\mu_k s \Longrightarrow \mu_k = \frac{v_b^2}{2gs}$$

$$\mu_k = \frac{(5\text{m/s})^2}{2(10\text{m/s}^2)(12.5\text{m})} \Longrightarrow \boxed{\mu_k = 0.1}$$

Problem 3 (20 points)

A uniform spherical shell of radius $r_s = 10\text{cm}$ can rotate about a vertical axis on the frictionless bearings. A massless rope passes around the equator of the shell, over a pulley of radius $r_p = 5\text{cm}$ and is attached to a small block. There is no friction on the pulley's axle and the rope does not slip on the pulley. The pulley and the small block have the same mass m and the sphere has mass $5m$. Find the angular speeds of the spherical shell (ω_s) and the pulley (ω_p) after the block has fallen 35 cm.



($I_{\text{sphere}} = \frac{2}{3}mr^2$, $I_{\text{pulley}} = \frac{1}{2}mr^2$, $g = 10.0\ \text{m/s}^2$)

Apply conservation of mechanical energy for all system:

$$\Delta K + \Delta U = 0 \Longrightarrow \frac{1}{2}mv^2 + \frac{1}{2}I_s\omega_s^2 + \frac{1}{2}I_p\omega_p^2 - mgh \Longrightarrow \frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{2}{3}5mr_s^2\right)\omega_s^2 + \frac{1}{2}\left(\frac{1}{2}mr_p^2\right)\omega_p^2 = mgh$$

Remember that: $r_s\omega_s = r_p\omega_p = v$

$$\Longrightarrow \frac{1}{2}mv^2 + \frac{5}{3}mv^2 + \frac{1}{4}mv^2 = mgh \Longrightarrow \frac{29v^2}{12} = gh \Longrightarrow v = \sqrt{\frac{12gh}{29}} = \sqrt{\frac{12(10\text{m/s}^2)(0.35\text{m})}{29}} \Longrightarrow$$
$$v = \sqrt{\frac{42}{29}}\text{m/s}$$

Therefore:

$$\omega_s = \frac{v}{r_s} = \frac{\sqrt{\frac{42}{29}}\text{m/s}}{0.1\text{m}} = 10\sqrt{\frac{42}{29}}\text{rad/s}$$

$$\omega_p = \frac{v}{r_p} = \frac{\sqrt{\frac{42}{29}}\text{m/s}}{0.05\text{m}} = 20\sqrt{\frac{42}{29}}\text{rad/s}$$

Problem 4 (15 points)

A 2.0-kg disk spins up from rest to its final angular speed with constant angular acceleration of 2rad/s^2. A point located 20.0 cm from the axis of rotation of the disk moves with a speed of 18 m/s when the disk is at full speed.

(a) (7 pts.) How much time it takes for the disk to reach its final angular speed?

$$\omega = \frac{v}{r} = \frac{18\text{m/s}}{0.2\text{m}} = 90\text{rad/s}$$

From rotational kinematics:

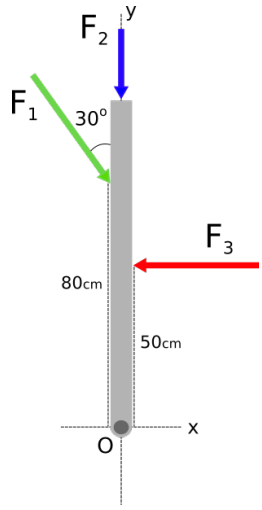
$$\omega = \omega_0 + \alpha t \Longrightarrow t = \frac{\omega - \omega_0}{\alpha} = \frac{90\text{rad/s}}{2\text{rad/s}^2} \Longrightarrow \boxed{t = 45\text{s}}$$

(b) (8 pts.) How many revolutions does the disk make as it goes from rest to its final angular speed?

$$\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2 = \frac{1}{2}(2\text{rad/s}^2)(45\text{s})^2 = 2025\text{rad}$$

$$\#\text{of rev.} = \frac{\theta}{2\pi} \Longrightarrow \#\text{of rev.} = \frac{2025\text{rad}}{2\pi}$$

Problem 5 (15 points)



A 300g meter stick at rest on a frictionless horizontal table is free to rotate around point O. Three forces $F_1 = 20\text{N}$, $F_2 = 18\text{N}$ and $F_3 = 15\text{N}$ are applied simultaneously as shown in the figure. ($I_{\text{rod}} = 1/3\, ML^2$)

(a) (6 pts.) Find the magnitude and direction of each torque

$$\vec{\tau}_1 = \vec{r}_1 \times \vec{F_1} = (0.8\text{m}(20\text{N})\sin 150°)(-\hat{k}) = (8.0\text{N.m})(-\hat{k})$$

$$\vec{\tau}_2 = \vec{r}_2 \times \vec{F_2} = (1.0\text{m}(18\text{N})\sin 0 = 0)$$

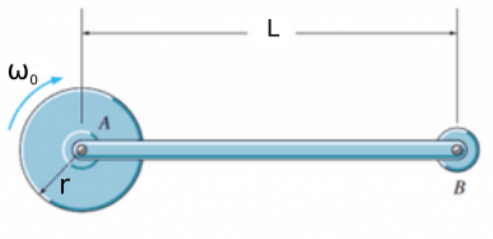
$$\vec{\tau}_3 = \vec{r}_3 \times \vec{F_3} = (0.5\text{m})(15\text{N})\sin 90°(\hat{k}) = (7.5\text{N.m})(\hat{k})$$

(b) (9 pts.) Find the angular acceleration of the stick.

$$\sum \vec{\tau} = I\vec{\alpha} \Longrightarrow \vec{\alpha} = \frac{\sum \vec{\tau}}{I} = \frac{\vec{\tau}_1 + \vec{\tau}_2 + \vec{\tau}_3}{1/3ML^2} = \frac{(-8.0\text{N.m})\hat{k} + (7.5\text{N.m})\hat{k}}{1/3(0.3\text{kg})(1\text{m})^2} \Longrightarrow \boxed{\vec{\alpha} = (-5\text{rad/s}^2)\hat{k}}$$

Problem 6 (20 points)

A uniform rod AB of length $L = 1\text{m}$ and mass $M = 3\text{kg}$ rests on a frictionless horizontal surface. The rod is pivoted about a fixed frictionless axis at one end (B) and it supports a disk of $m = 1\text{kg}$ with radius $r = 20\text{cm}$ at its other end (A). If the disk is given an angular velocity $\omega_0 = 100\text{rad/s}$ while the rod is held stationary and then released, determine the angular velocity of the rod after the disk has stopped spinning due to frictional resistance at bearing A. ($I_{\text{rod}} = 1/3\, ML^2$, $I_{\text{disk}} = 1/2\, mr^2$)
Hint: You may think parallel axis theorem



Apply conservation of angular momentum: $\vec{L}_i = \vec{L}_f \Longrightarrow I_i\omega_i = I_f\omega_f$

$$I_{(\text{d about A})}\omega_0 = I_{\text{r}}\omega + I_{(\text{d about B})}\omega \Longrightarrow \left(\frac{1}{2}mr^2\right)\omega_0 = \left(\frac{1}{3}ML^2\right)\omega + \left(\frac{1}{2}mr^2 + mL^2\right)\omega$$

$$\left(\frac{1}{2}(1\text{kg})(0.2\text{m})^2\right)(100\text{rad/s}) = \left(\frac{1}{3}(3\text{kg})(1\text{kg})^2\right)\omega + \left(\frac{1}{2}(1\text{kg})(0.2\text{m})^2 + (1\text{kg})(1\text{m})^2\right)\omega \Longrightarrow \boxed{\omega = \frac{1}{1.01}\text{rad/s}}$$